

I&C BEOC COOLING CONTROL SPECIFICATION & DESIGN

GA PROJECT 30424

REVISION HISTORY

Revisio n	Date	Description of Changes
A	See Watermark	Initial Release

POINT OF CONTACT INFORMATION

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1 INTRODUCTION

1.1 Purpose

This document outlines the Instrumentation and Controls design for the interior air cooling of the Back End Optics and Cameras (BEOC). The objective of the document is to define the system architecture, the main components related to I&C, control scheme for cooling of BEOC, the associated alarms and faults, and present important software functionality.

2 SUBSYSTEM OVERVIEW

2.1 BEOC Cooling Control System Overview

The BEOC subsystem is located in the Port Cell, and encloses components that need to be in a temperature-controlled environment. The IR Optical Bench, Visible Optical Bench, RF active matchers, and other components as shown in figure 1, and all need to operate below a maximum operational temperature, and for optimum performance, temperature fluctuations and gradients need to be minimized.

Description	Power Dissipation (W)	Total Power (W)	Power in	Op Temp. Non-op Temp.	Relative Humidity (%, Non Condensing)
IR Camera	38	76	24 VDC, or 100-240VAC	Op: -20 to 50	-
VIS Camera 1	9	9	9W, 12 V	Op: 5 to 65 Non-Op: -10 to 70	10 to 90
VIS Camera 2	10	10	10W, 12-24 VDC	Op: 0 to 60 Non-Op: -10 to 70	10 to 90
RF Matching Box	27	54	90-264 VAC 50/60 Hz, 90 VA (0.7 PF)	Op: 5 to 35 Non-Op: -30 to 60	30-80

The BEOC cooling is accomplished with a liquid-to-air heat exchanger, thermocouples for temperature monitoring, and a variable speed cooling fan. The Slow Controller Programmable Logic Controller (PLC) in the Diagnostic Hall will monitor the thermocouples and determine the speed of the cooling fan. The primary components of the BEOC cooling system are,

- A liquid to air heat exchanger and
- A variable speed cooling fan.
- Thermocouples
- The Slow Controller PLC in the Diagnostics Hall
- PLC thermocouple input modules (AI module)
- A PLC fan speed control module output module (PWM module)

The camera cooling control should be able to maintain the ambient temperature inside the BEOC without the 20 deg-25 deg range with minimum temperature gradient. The IR camera life is extended if cameras are maintained at stable temperatures.

All the BEOC cooling components controlled by Siemens PLC located in the diagnostic hall cubicle. The location and cabling are shown in P&ID and CBD, respectively.

2.2 System Interfaces

The chilled water to the heat exchanger is provided by ITER PBS 26.CH (Chilled Water Supply). The chilled water is fed to BEOC heat exchanger, and it is assumed PBS 26.CH monitors the temperature and controls the water pressure upstream of BEOC. This design assumes a chilled water with minimum fluctuation in temperatures. BEOC cooling design currently does not have temperature or pressure request to PBS 26.CH planned.

2.3 Special Design Considerations

Being in the Port Cell, the BEOC is in an area that is exposed to neutron, gamma, and electromagnetic radiation. The BEOC is surrounded with protective shielding layers to absorb this radiation and protect internal components. The heat exchanger is mounted to the outside of the BEOC, and therefore outside the protective shielding. To mitigate the effects of radiation, a modified commercial-off-the-shelf (COTS) heat exchanger is planned based on prototype activity performed by EPSYL and IO together and is detailed report ITER_D_DJDXCL.

A modified version of Pfannenbergl heat exchanger model PWS 7062, 0.6KW power is selected for UWAVS based on ITER_D_DJDXCL. The modifications include the removal of a solenoid valve, relocating a connector, and changing the fan motor from an AC to a DC motor. The distance between the PLC in the Diagnostic Hall and the BEOC, when taking cable routing into account, can be approximated to approximately 200 meters in case of furthest port #2 and pot #17. This distance is excessive for typical cable motor control and thermocouple cables. The PLC and PWM module is baselined in Diagnostics hall due to proclivity from not placing electronics equipment in BEOC / port cell. Any risk to driving DC motor over long distance will be tested and addressed during MRR / Manufacturing phase.

2.4 Design Rationalization

The I&C design rationalization considers that successful operation of the UWAVS, and ultimately the reactor itself, relies on many complex systems and subsystems working as expected. All the systems and subsystems need to work in concert with each other to achieve successful operation. The design of the UWAVS should minimize the chances of a failed diagnostic measurement. At the same time, operation of UWAVS should be reliable, intuitive, and simple for an operator who may be simultaneously overseeing the operation of many systems and subsystems. The cooling water for the BEOC is supplied by the PBS 26.CH, and the water temperature and flow is controlled by these upstream systems.

The COTS version of the planned heat exchanger utilizes a cooling water solenoid to control the flow of the cooling water. This on/off control of cooling water would result in temperature oscillations around a temperature set point. Additionally, this solenoid is in the heat exchanger, outside the BEOC. This would expose the solenoid to radiation discussed earlier and require shielding and/or relocation of the solenoid. As an alternate design, the solenoid is removed, and water flow is not controlled by the UWAVS. Instead, cooling water flow is continuous, and cooling is controlled using a variable speed fan inside the BEOC to control the amount of air flowing through the heat exchanger. This eliminates the need to relocate or shield the solenoid and minimizes temperature variation and gradients inside the BEOC due to solenoid cycling. Additional alteration of the heat exchanger design from the COTS version is changing the motor from an AC motor to a DC motor. This simplifies the hardware needed for variable speed control of the motor.

Based on analysis of mechanical team for BEOC cooling, the Pfannenberger model PWS 7062 at 0.6kW capacity was found adequate to cool the area and hence selected by I&C team. **The thermocouples are**

2.5 Calibration requirements at FAT / Commissioning / STM / LTM

At commissioning phase, it is recommended to validate the temperature zones specified by 55.GA for control strategy. At commissioning the temperature of BEOC can be studied when IR cameras are on to validate the temperature control feedback loop to vary the fan speed.

Also, the temperature of BEOC can be studied when RF active matchers are on in BEOC and validate temperature to fan speed control loop. The alarming and monitoring strategy can make more sophisticated based on the understanding.

3 SYSTEM ARCHITECTURE & DESIGN

3.1 Temperature Control and Monitoring

Multiple temperature points are monitored in, on, or around different components of the BEOC. These include the IR Optical Bench, the RF matchers, and other points in the BEOC structure and optics. **These are called out in (TODO: Reference P&ID Drawing?** Is this the controlling document for what temperatures are being monitored?). **TODO:** There are no MTE temperature elements listed in the visible optical bench on the P&ID. Is this area part of the temp control/monitoring, and how is temperature

obtained? A monitored temperature may be used as a temperature control point, or it may be simply used for monitoring and alarming and not be included in the temperature control scheme. Depending on the operational mode of the system, the monitored temperatures used for control may change. In normal operation, the critical equipment is the IR and visible cameras, for example. While calibration is occurring, the calibration equipment is critical. (TODO: This statement needs to be checked. Do we know enough or should we have a table of modes vs ctrl temps?)

3.2 Temperature Control Scheme

The available temperature measurements are considered for validity. Any that are out of range, broken, disconnected, marked to be ignored, or invalid for some reason, are ignored with respect to temperature control. Related temperature measurements are broken into “zones”. Valid temperature measurements within a zone are averaged. A zone shall be considered a “control zone” depending on the operational mode of the system. If a zone is a control zone (being considered for control), then its average is considered in the control scheme. With an average for each control zone, the Process Variable (PV) used for temperature control is the Maximum Value of all the control zone averages. This calculated PV will be compared to the Set Point, and a Proportional Integral Derivative (PID) control loop will be used to control fan speed.

3.3 Temperature Zones

TODO: List zones and measurement points here, like a tabular version of the P&ID?

3.4 Fan Speed Control

Variable fan speed is used to control the amount of air blown through the heat exchanger and thus is the Controlled Variable (CV) for BEOC Cooling control. The heat exchanger fan being considered is a 24 VDC, 48 Watt fan. The power to each fan is controlled by the PLC via a DC Digital Output Module configured for Pulse Width Modulated (PWM) control. By varying the duty cycle of the PWM signal, the fan speed is controlled. The PID loop used to control temperature will output a fan speed demand from 0% - 100%, and this demand is directly translated into duty cycle. This module resides in a local or remote I/O rack in the Diagnostic Hall, and a single module can control power for up to 8 fans. Fan cables are to be run from the Diagnostic Hall to the BEOC for each fan.

Note there will likely be a minimum non-zero percent corresponding to a minimum speed that will be needed to run the fan. Additionally, the heat exchanger may only be effective with some minimum amount of air being blown through it. These factors mean the fan will likely be turned off if demand is below some non-zero minimum demand.

3.5 Control strategy & Operational Scenarios

BEOC cooling is operated continuously throughout lifetime of 55.GA. The controlling “zones” are based on operational scenario of 55.GA. During pulsed operation, the zone near IR camera is selected for

control. During mirror cleaning, which happens in STM, the temperature zones near RF active matchers are selected.

3.6 Technical Specifications

The Pfannenbergl PWS 7062 Specification shown in figure 2.

Data		PWS 7062	
Rated voltage $\pm 10\%$		50/60 Hz AC	60 Hz AC
		230 V	115 V
Cooling capacity at 200 l/h	W10/A35	600 W	
Ambient temperature		+1 ... +70 °C / +34 ... +158 °F	
Water inlet temperature		+1 ... +35 °C / +34 ... +95 °F	
Degrees of protection according to EN 60529	IP 55	towards the electrical enclosure if used as intended by the manufacturer	

3.7 Exception Handling, Faults and Alarms

Each zone has multiple temperature monitoring points. If one or more is not working or is giving an out-of-range signal, then it is ignored for the purposes of temperature control. An "Alarm" is a condition that alerts an operator of an abnormal or deteriorated condition, and which limits operation, and generally requires intervention. A "Warning" is a condition that alerts an operator of an abnormal or deteriorated condition, but which may not limit operation, or have only a minor impact on operation, and requires little or no intervention.

In MRR, use of DC motor circuit open can be detection using current sensing device interfacing to PLC can be evaluated to detect C motor failure for certain.

Fault Type	Description	Possible Cause	System Response	Error Recovery
Warning	Temperature invalid	Sensor problem	Annunciate warning	Fix sensor or wiring
		PLC Module channel problem		Fix channel configuration
		PLC misconfigured		Fix the configuration
Warning	Zone High	Heat load is high	Annunciate warning	Lower the heat

	Temperature Warning			load
		Cooling is insufficient		Investigate fan or air flow
		Fan or air flow problem		
Alarm	Zone High Temperature Alarm (Operation may be affected)	Heat load is high	Annunciate alarm	Lower the heat load
		Cooling is insufficient		Investigate fan or air flow
		Fan or air flow problem		
Warning	Temperature outlier (valid but statistical outlier) (point)	Sensor problem	Annunciate warning	Investigate sensor and wiring
		Sensor placement or location problem		Investigate sensor location
Alarm	Zone No Valid Temperature	All zone temperatures are invalid	Annunciate alarm	Fix sensors/wiring
		Module broken, missing or misconfigured	Zone temperature will be ignored	Fix or install the module
		PLC misconfigured		Fix the configuration
Alarm	Motor Over Current (May lead to loss of cooling)	Wiring issue	Annunciate alarm	Fix wiring
		Motor fault		Investigate motor operation
Alarm	Motor Open Circuit (May lead to loss of cooling)	Wiring issue	Annunciate alarm	Fix wiring
		Motor fault		Investigate motor operation

4 TESTING

Please provide how the subsystem can be tested for hardware and software to show successful design.

1. Hardware test plan
2. Software test plan

5 RISKS

Please provide risk assessment if any.

6 BOM

Device	Manufacturer Number/ ITER Part Number	Quantity
Heat Exchanger (modified)	Pfannenbergs PWS 7062	1
Cooling Fan	Ebmpapst GmbH RER 175-42/14/2 TDMLP	1
Slow Controller PLC	Siemens 6ES7516-3AP03- 0AB0	1
Mounting Rail	Siemens 6ES5710-8MA11	1
Digital Output Module	Siemens 6ES7522-5BF00- 0AB0	1

7 OBSOLESCENCE MANAGEMENT

Please provide any vendor details that support long term support for selected components.

8 HMI VARIABLES

1. Trend pop ups (T,Rh,Press)
2. Graphic of beamline showing valve conditions and sensor readings